

2.5 - THE GRID WE INHERITED

2.5.1. On the bridge of the Titanic

It is Sunday evening, late March 2026 when I decide things are so wrong that I have to take a stand and say it clearly: we must break up the electricity grid.

To almost any grid expert I know this sounds like heresy, but if we don't it will ruin us financially, keep us chained to fossil fuels, and make us increasingly vulnerable.

I am at home finishing a report for the Dutch government that I've been working on – on and off – for over a year that answers the question: to what extent could local autonomy solve the huge grid congestion problems the Netherlands is facing?

I take a break and open Bluesky to see what is going on. Someone alerts me to the fact that the Dutch grid operators have just published their new joint cost estimate for the onshore electricity grid through 2040: [FIEN26](#). It is the fourth in a series.

As I start reading the cost estimate I can hardly believe my eyes. Since the [previous estimate](#), just 15 months ago, the estimated cost of reinforcing the grid has risen by roughly **€5,000 per household**. This on top of the roughly **€35,000 per household** that were already estimated. Not for solar panels. Not for a heat pump. Not for insulation. Just for *extra* wires that connect our sources to our users of electricity. And on top of that the Netherlands loses economic growth worth at least **€1,000 per household per year** because of businesses and households that [cannot be connected to the grid](#).

My first emotion, honestly, is despair. How are poor households going to afford this? Will this block the transition away from fossil fuel?

I talk to my colleagues at Zenmo. We quickly agree that our models indicate this is a spectacularly expensive way to do it. A different architecture — local coordination, flexible demand, smart use of batteries — could deliver the same transition at roughly half the cost, with no waiting list and no yearly economic damage. My colleagues are resigned. We are small. Grid operators are large. It will go their way. "That's just the way it is." But in my head I hear I hear Bruce Hornsby singing the next line: "Ah, but don't you believe them."

Then I call the people I know who work at municipalities, provinces, and consultancies. They are uniformly outraged. [Roosmarijn Sweers](#), who handles energy planning in the [Drechtsteden](#) region south of Rotterdam, and Bastiaan de Jonge at the province of [Utrecht](#). Both sketch the same picture: if this is really how it is going to go, the consequences for municipalities and provinces will be devastating. Five to ten years of effective lockdown, during which plans for siting new businesses and building new houses will be largely grounded. For them, this is a nightmare scenario. And my friend [Maya van Steenhoven](#) has of course already read it and calculated the costs per households. "*How are ordinary households supposed to pay this kind of bill?*" she furiously exclaims with the righteous anger of someone who gives a damn.

The next day I happen to attend a workshop with the people who actually wrote FIEN26. Their reaction is not outrage. It is not panic. They are *proud*. They have improved their methodology. Yes, the number is higher, but the estimate is more precise — that is progress. This is the best cost estimate they have ever produced.

They are on the bridge of this ship, and on the bridge nobody is talking about a course correction as the Titanic is about to run into the iceberg. Instead they ask themselves *how do we reinforce the hull enough to crush this iceberg*. The Titanic is being redesigned into an icebreaker — a technically possible project, but at monumental cost. A course correction is simply not on the menu. To understand how the hell that is possible, we have to take a step back.

2.5.2. The greatest invention of the twentieth century

In 1745, a Dutch professor named [Pieter van Musschenbroek](#) discovered that you could store electricity in a glass jar. He called named the device -- after the city he lived in -- a [Leyden jar](#). His first attempt to charge it knocked him across the room. He wrote to a French friend that he would not repeat the experiment "for the whole of France." That friend, [Abbé Jean-Antoine Nollet](#), of course replicated it immediately and within a year was standing before King Louis XV at Versailles with [180 Royal Guards in a circle holding hands](#). As he sent a charge from a Leyden jar through the whole line, all soldiers leapt into the air in unison. The king requested a repeat. In Paris, Nollet did it again with a line of Carthusian monks, over a kilometre long.

For the next century and a half, electricity lived mostly in jars and in bodies, and in the imaginations of a handful of people who sensed that something new was happening. In 1831, Michael Faraday built the first generator in his basement at the Royal Institution — a copper disc spun between the poles of a horseshoe magnet — and produced, for the first time in human history, a continuous electric current on demand. In the 1860s, James Clerk Maxwell wrote down the four equations that describe every electromagnetic phenomenon in the universe on a single sheet of paper, which is one of those events that retrospectively seems much too large to have happened even in a single decade. But electricity, for almost everyone alive in the 1870s, remained a curiosity. You read about it in the papers. You had never seen it.

And then, rather suddenly, you had. At three o'clock in the afternoon of 4 September 1882, in a [four-storey brownstone at 255–257 Pearl Street in lower Manhattan](#), Thomas Edison turned on the first electricity grid. Steam engines started to turn six 27-ton dynamos and four hundred electric lamps over a square mile of lower Manhattan turned on. Edison's main investor Morgan and his colleagues were there. They had, until that afternoon, lived their entire working lives by candle, gaslight, and the [occasional, brutal glare](#) of an arc lamp on a boulevard. The *New York Times*, also experienced the miracle and the next day described the miracle as *soft, mellow, and graceful to the eye ... without a particle of flicker to make the head ache*.

Within a year, Edison had a dozen stations in operation. Within a decade, every major city in the United States and Europe had central stations of its own. Electric streetlights replaced gas lamps along the grand boulevards of Paris and Berlin and Budapest, and

people who had spent their lives moving indoors at sunset discovered that they could now walk outside at ten at night as easily as at ten in the morning. And it was not just lamps. Factories installed electric motors and doubled their output without adding a single worker. Trams ran on overhead lines and displaced entire generations of horses — a huge, unnoticed, and enormously consequential environmental improvement, since a city of a million people had previously also been a city of a hundred thousand horses, with all the manure that implies. Electric elevators made twenty-storey buildings feasible and, in doing so, invented the skyline.

Edison's first grid ran on direct current, which turned out to be a technological dead end. Direct current could not be transformed — it could not be stepped up to the high voltages required to move electricity across long distances without losing most of it as heat in the wire. Alternating current could. And so, over the course of the 1890s, alternating current — championed by [Nikola Tesla](#) and [George Westinghouse](#) — won the [War of Currents](#). And the AC grid then spread, within a single century, across North America, Europe, Japan, the Soviet Union, and beyond.

Today, the grid holds tens of thousands of generators in synchrony across entire continents. A nuclear reactor in Finland, a hydroelectric dam in the Spanish Pyrenees, a lignite plant in Poland, a gas turbine in the Netherlands, a wind farm in the North Sea — every one of them is spinning at exactly the same rate, because every one of them is connected to the same continental-scale alternating-current machine. Supply and demand are balanced every second of every day. The electricity grid keeps billions of people warm, fed, entertained, and alive in a way unconceivable two centuries earlier.

The [National Academy of Engineering](#) named the electric grid the greatest engineering achievement of the twentieth century — ahead of the automobile, the aeroplane, the computer, and the internet. I worked at a grid operator for a number of years and I think most of us agreed with that. The grid truly is the greatest machine humanity ever built. The people on the bridge want to preserve this marvel.

2.5.3. The grid is the perfect answer to a question that is no longer being asked

There is a reason the grid is the way it is. When it was built, a few things were clear to anybody designing it.

Fuel was costly. Coal had to be mined, transported, and burned. Bigger coal fired power plants did that more efficiently and so [economies of scale](#) dictated that bigger was better. As technology improved and size increased, thermal efficiency climbed from somewhere around 10% in 1900 to about [45% in the best modern plants](#). Bigger meant cheaper per kilowatt-hour. Big centralised plants also meant fewer people were needed to manage them, which brought the cost down further. On top of all that, burning coal was dirty, so you did not want the plant near where people -- especially rich people -- lived. All this meant that power from the plants had to travel long distances. Long distance cables meant high voltages — the only way to move serious amounts of electricity without wasting most of it as heat in the wire. High voltages, in the pre-semiconductor era, meant alternating current.

Storage was impractical. Batteries existed, but they were far too expensive to store electricity at any useful scale. This meant that supply had to chase demand, every second of the day. The continent needed a universal language of balance, and it found one: frequency. If generation keeps up with demand, the grid runs at 50 Hz. If demand suddenly outpaces generation, the spinning turbines slow a little and the frequency drops. If generation outpaces demand, frequency rises. Every generator on the continent listens to frequency and adjusts. The inertia of all those spinning masses acts as a buffer: during a sudden disturbance, the rotating metal buys the operators a few seconds to respond before the system drifts. This is how grid operators keep the grid stable.

Computers were not available. For the first decades of the grid there were no computers at all, and for the decades after that they were rare and expensive. So you could not put intelligence at every node. You ran the grid, instead, from a control room, with human planners, organised hierarchically. Wholesale markets were laid on top — day-ahead, intraday, balancing — to allocate generation between competing plants ([EPEX Spot](#) became the reference for Europe). Each participant had to declare in advance what they would produce or consume; a system of "balance responsible parties" with fines for deviation kept the whole thing tractable. The lack of computers meant strict centralized planning.

Consumers had nothing to contribute. In the era that built the grid, the ideal of industrial efficiency was the assembly line. Cheap workers adapted to expensive machines, not the other way round. Consumers were treated the same way: predictable, passive, aggregate-able. Your kettle boils, your fridge hums, your light comes on. You just do what you always do, which makes you predictable, especially in large numbers. That way planners could treat demand as a static statistical pattern. Your inability to adjust was not a bug but a feature. The grid depended on it.

The grid was a copper plate. From all of this came the crowning promise: wherever you are and whenever you want electricity: you just throw a switch and you will have it. Everyone understand the allure of that simple promise.

A few simple truths that made the grid the way it is. And yet, quietly and without ceremony, technological progress has turned all of these unalienable truths into untruths.

2.5.4. A New Era is Dawning for the Grid

These are the new truths that we should build the new grid on:

Solar and wind are abundant, but arrive whenever they feel like it. Sunlight and wind are already cheaper than coal overall, but unlike coal, they do not take orders. You cannot call up the sun and ask it to work harder on Tuesday. From the point of view of the old paradigm, this is a catastrophe — the planning apparatus built around predictable fossil dispatch is in complete disarray. The fossil industry has responded, reasonably from its own point of view, by rebranding itself as "firm" and "dispatchable" — coded language for *reliable, unlike solar and wind*. But from the point of view of those who understand how the new grid can actually work, they are simply denying themselves abundance.

Storage is not just possible — it is getting cheap. Battery costs have fallen on a learning curve steeper than solar's. Stationary batteries plus the batteries sitting under electric cars will, within the decade, represent [roughly ten times the balancing power](#) that thermal generation ever had — and unlike a coal plant, a battery can respond in milliseconds. And yet the grid operator today does not use them for balancing. The concept of a self-balancing grid is still too alien.

Computation is becoming essentially free. In 2025, a microprocessor with the capabilities of a 2005 supercomputer costs less than a sandwich. And yet the grid, at its coordination layer, is still run broadly the way it was run in the 1970s, with the addition of some SCADA screens and a great deal of PowerPoint.

Consumers are active. Remember the assembly-line deal from the previous section — the one where the consumer's job was to be predictable? That deal is over. Every rooftop with solar, every EV on the driveway, every heat pump in the utility cupboard, every home battery in the garage: these are not predictable loads. They are active participants. People want control of their own equipment. Tell them when they can charge their car, and they will politely tell you where to put that idea. But ask them nicely to actively solve grid problems in return for lower grid costs and they are eager to help.

The grid is no longer a copper plate. The promise that location and time did not matter cannot be sustained in a world where generation is distributed, demand is flexible, and the direction of flow changes with the weather. The result of trying to pretend the copper plate still exists in the Netherlands is that congestion is everywhere and connection queues are measured in years. And yet the Dutch grid has on average [80% of physical capacity left](#).

The way we are trying to solve congestion problems — by throwing ever more copper at it — is [not working](#). I've come to the conclusion that the real scarcity is not the amount of copper that is available, but the imagination by which it is utilized.

Allow me to summarize what is happening in the table below.

The first column is the grid we built. The second column is where we are now: grid operators pretending the first column still applies while the world has moved on. The third column is we are going.

Four years ago, the third column was little more than a premonition for me. An intuition if you like. I hadn't yet discovered the visionaries that had already described it. I didn't know yet that that it was business as usual in other domains. But right now, after years of simulation, fieldwork, and just reading the ruinously expensive new cost estimate of the grid operators, I'm willing to die on this hill: we need to reorganise the grid using decentralized semi-autonomous microgrids. They will make the grid cheaper and more efficient while solving congestion.

But there is another – arguably even more important reason – to break up the grid into these decentralized semi-autonomous microgrids, and that is to make it less fragile.

2.5.5. The Current Grid is Fragile by Design

To explain why the grid should be split up instead of being one big monolithic system, it is instructive to look at potatoes in Ireland.

To explain why the grid should not be one big monolithic system but a loosely coupled ecosystems of microgrids, it is instructive to start with the story of the potato in Ireland. The potato reaches Ireland in the seventeenth century and spreads as food for the rural poor over the following hundred years. It yields more calories per acre than any grain and grows in soil that would barely sustain oats. By 1800 it is the staple of about half the population; by 1840, the near-exclusive staple of the poorest third. Irish population rises from two million in 1700 to eight million in 1840, essentially on the back of this one crop. Within potatoes, a cultivar called the Lumper emerges as the highest-yielding variety. By the 1840s, in much of Ireland, this is the variety that gives everyone food on the table.

In September 1845, *Phytophthora infestans*, a pathogen that has jumped the Atlantic on a ship, reaches Belgium, then Flanders, then Ireland. It immediately starts attacking local European potato's. My country the Netherlands is hit hard and over 50 thousand people die. But because there is a large variety of potato species in the Netherlands, because poor people also eat bread from wheat and rye, because many families do more than work on the land, and because the government intervenes, the size of the famine is limited. In Ireland the effects are far more severe. Because the Lumper is very susceptible to the pathogen and because it's the only type of potato grown in Ireland, counties watch their entire harvest turn black. In 1846 the pathogen attacks again, and in 1848 again. Because it's the only staple poor people survive on, the end result is one million dead of starvation and disease. One million emigrated, mostly to the United States. Ireland's population drops by a quarter in seven years and still has not recovered to its 1845 level in 2026. Genealogically, it is the largest single event in Irish history. The British government's response — insistence on free-market doctrine, continued grain exports out of Ireland during the famine — makes the outcome worse. But the catastrophe was built in before any politician opened their mouth. It was rooted in a focus on efficiency over resilience.

Another example of how vulnerable homogeneous systems are can still be tasted when you drink European wine. Every wine grape in Europe is the same species, *Vitis vinifera*. When an American botanist imported experimental vines to Languedoc in the late 1850s; he also imported a tiny aphid on its roots. These aphids were harmless to the American vines with their hardened roots but European vines have no defence. By 1889, French wine production has halved, 40% of vineyards are dead, and thousands of vintners have emigrated. The eventual fix was to graft every European vine onto American rootstock that developed natural resistance to this pathogen.

Or take the American dust bowl. First World War wheat prices, tractors, and combines combine to tear up the prairie of the Great Plains. The deep-rooted native grasses that had held soil together for ten thousand years are replaced by wheat with roots twenty centimetres deep. When drought arrives in 1931, there is nothing holding the topsoil in place. On black Sunday, 14 April 1935: a thousand-mile dust storm turns afternoon to midnight. By 1938, 75% of topsoil is gone from half a million square kilometres (fifteen

times the size of the Netherlands) and 3.5 million people are displaced. Another human system optimised for short term profit suffers because it's not resilient.

Modern mega-wildfires vividly illustrate the same phenomenon at work. A century of fire suppression has converted forests that have burned in small patches every decade for ten thousand years into even-aged, uniform, heavy-fuel landscapes that burn all at once when suppression inevitably fails, especially in the age of global warming. In California in 2018, the town of Paradise is destroyed in a morning. In 2019-2020 Australia faces it's Black Summer: 100 thousand square kilometres burned, and an estimated three billion vertebrate animals killed or displaced.

Fragility of human designed systems is not just about natural system. Other examples include the 2008 financial crisis, the Boeing 737 MAX crashes or the many examples in software. Some of you might remember CrowdStrike that crashed every Windows machine receiving it's software update, leaving many hospitals, airports, banks, stock exchanges and newsrooms in disarray but I think M.E.Doc is even more instructive. M.E.Doc is used by almost every business in Ukraine for tax compliance. In 2017 Russian state hackers manage to plant a backdoor in an automatic update and installed NotPetya: wiper software disguised as ransomware. It caused worldwide destruction. Maersk, the world's largest shipping company, had M.E.Doc installed on one machine in its Odessa office. Within hours, 45,000 PCs, 4,000 servers, and 2,500 applications across 600 offices and 76 ports were dead. The global supply chain ships 20% of its containers on Maersk vessels; there is now no IT. The company recovers only because of a single backup in Ghana that happened to be offline during a local power cut on the day of the attack — the one thing that survived was the one thing not fully integrated.

Let's finish this tour of destruction caused by fragile human designs with a grid blackout: the Iberian blackout, 28 April 2025. After one solar plant goes offline, voltage oscillations generate a cascade with many other solar systems going offline when voltages exceed limits at which they are permitted to function. It leaves 55 million people without power across Spain and Portugal. Trains stop in tunnels. Traffic lights die. Hospitals go generators (that thankfully all work). On social media, the first response is to blame solar and the lack of fossil (or nuclear) inertia. Later [ENTSO-E's final report](#) paints a very different picture, blaming mainly the regulatory framework that forced solar systems to turn off at the same time. As we will see, a far more robust defence would have been to allow solar systems to stabilize the voltage independently, instead of ordering them to shut off once the big fossil plants cannot manage anymore. And if we add smartly controlled batteries to the mix, blackouts like these become virtually impossible.

All these examples show how we invite disaster if we allow tightly coupled homogeneous systems without buffers or internal boundaries. The grid is a prime example of a system inviting disaster this way. Nature shows us there is a better way.

2.5.6. We Should Learn from Nature

Life on Earth has survived worse than anything humanity has experienced. The [Permian-Triassic extinction](#) 252 million years ago killed about ninety-six per cent of marine species and seventy per cent of terrestrial vertebrates — the closest life has ever come to packing it

in. The [Cretaceous-Paleogene asteroid](#) 66 million years ago took out three-quarters of all species, including every non-avian dinosaur. [Snowball Earth](#) froze the planet to the equator for something like ten million years, possibly twice. Any single one of these events would have ended human civilisation as we currently have it arranged. For the biosphere, none of them did. Something about the way life is put together allowed it to take hits on a scale that would turn human civilization to dust. What is it?

One thing life does, which our engineered systems almost never do, is nest smaller semi-autonomous systems inside larger ones while keeping the smaller ones alive. One of the most extreme examples is happening inside your cells right now. Roughly [two billion years ago](#), a single-celled organism swallowed another single-celled organism and, for whatever reason, did not digest it. The swallowed bacterium could use oxygen to generate energy. The host could not. They got on. Over generations they became a partnership: the host kept the bacterium fed and protected; the bacterium kept the host powered.

That swallowed bacterium is the mitochondrion. Every one of the roughly [thirty trillion cells in your body](#) contains hundreds to thousands of its descendants. They still divide on their own schedule, on their own timing, and they still carry their own circular bacterial DNA, separate from the nuclear DNA you inherited from your parents. A [similar event later, separately](#) produced chloroplasts which allows leafs to turn sunlight into energy: because of a merger, two billion years old, between a larger cell and a smaller one that was already good at turning light into sugar. You, the reader, are a collaboration of absorbed bacteria, all the way down and so is the spinach in your fridge.

When something goes wrong inside a mitochondrion, the cell it sits in usually carries on. When something goes wrong with one cell, the tissue it sits in carries on. When something goes wrong with one tissue, the organism carries on. Damage at one level does not cause catastrophic cascades up or down the chain. The whole reason you can survive a cut, a fever, a minor infection — the reason any complex organism survives anything — is that life has for billions of years run on nested, semi-autonomous systems with their own energy, their own repair mechanisms, and their own ability to keep going when a neighbour fails.

Life has even learned to *profit* from disaster. In the summer of 1988, Yellowstone caught fire. A drought, dry lightning, and gale-force winds combined, and by the end of the season [about thirty-six per cent of the park had burned](#) — a quarter of a million hectares. It seemed a long term disaster but within a decade the forest was denser than before, because the dominant tree — the lodgepole pine, *Pinus contorta* — produces cones sealed with resin that open only above about fifty degrees Celsius. When the fire kills the parent tree, the cones crack and drop tens of thousands of seeds onto freshly fertilised soil with no competition. The fire does not destroy the pine's reproductive architecture. [It activates it.](#)

2.5.7. The course correction won't come from the bridge

The current crew and captain are feverishly turning the Titanic into an icebreaker. They are trying to create a grid for the future based on outdated thinking. And if given an unlimited budget they will probably succeed eventually. But why? If you look under the hood, the grid with all its separate transformers is already a system of potential microgrids. A flotilla masquerading as big ship. Why not allow each ship the freedom to steer around icebergs on

its own? Why not when this is what gets us to a future of clean abundant energy? Why not when this flotilla of independent ships gives us more safety and speed at lower costs?

This section I honestly found hard to write. As someone who has worked at the grid operator for a number of years I want to be fair to my former colleagues.

I went to Alliander, the largest DSO in the Netherlands, because I thought improving the electricity grid was the most important thing I could work on. If you want to decarbonise the economy, you have to electrify almost everything, and if you have to electrify almost everything, the grid is where the project succeeds or fails, I thought. I wanted to understand and fix the problem from the inside. That was the way I envisioned having the most impact.

Dutch grid operators are [government-owned](#). Their objectives should be aligned with those of society at large. There are no shareholders extracting value in ways that might be at odds with societal goals. And the resulting grid is one of the most reliable in the world. If change from within was going to work anywhere, Alliander was the grid operator where it should have worked.

I left Alliander because I became convinced that change from within was simply not an option. The engineers inside Alliander had plenty of ideas. Many of them had better ideas and deeper knowledge than I had. But the system they worked inside could only accommodate so much change. The big course corrections that were needed were simply not on the menu.

And to the extent course corrections were on the menu inside the grid operator, they were prevented because of the gridlock that existed in a system that also contains the regulator ([ACM](#)), and the ministry ([KGG](#)). Any change each of them makes has repercussions for the others in an elaborate system that rewards continuity and punishes deviation. The literature on regulatory capture and on [institutional path dependency and inertia](#) in energy infrastructure describes this pattern with alarming accuracy.

This in turn is part of a larger pattern on [monopolies and innovation](#) that is documented at length in the scientific literature and that shows that it's exceptional for market leaders to disrupt themselves, even if their survival depends on it. For monopolies, disruption is almost impossible.

So that is why this book is written for a broad audience. Ten years ago I would have written it for grid operators since they are the experts on the subject, but now I understand that they are captured inside a system that cannot disrupt itself. It is up to you, dear reader, to provide the disruption. It's up to you to gain entrance to the bridge and demand a course correction, even if the crew tries to deny you access and even if the captain is unwilling to comply.